Hybrid Deep Learning and Quantum Machine Learning Approach for Accurate Polyp Segmentation in Colonoscopy Images

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***Abstract*—Colorectal cancer is a cause of cancer deaths worldwide. It causes deaths every year. Finding and removing polyps early through colonoscopy imaging is key to reducing deaths and improving survival rates.**

However looking at colonoscopy images manually is hard takes a lot of time and can be wrong. This is often because polyps come in sizes, shapes, textures and lighting conditions. These challenges show that we need computer systems that can help doctors diagnose accurately and reliably.

In this study we suggest a system that combines deep learning and quantum machine learning. This system is for accurately and quickly finding polyps in images. It uses a U-Net, a type of network good for analyzing medical images to find features in colonoscopy images.

To make the system better at finding polyps we add a layer that uses quantum computing. This layer helps find patterns that regular computers might miss.

We tested the model on a set of colonoscopy images with polyp masks. We used techniques to prepare the images and make the model work better on different images. The results show that the model works well with a Dice score of 0.9161, an Intersection over Union score of 0.8701 and a pixel accuracy of 0.9874.

These results mean that the model is good at finding polyps and works well on images. Using machine learning with deep learning not only makes the model better at finding polyps but also helps it work well in hard situations.

This new approach is a step, toward creating intelligent healthcare systems. It could help doctors diagnose patients in time making it easier for them to make decisions and improve patient outcomes.The use of quantum machine learning with learning makes segmentation more precise. It also helps the model work well in situations.

This mix of quantum machine learning and deep learning is a start for creating future intelligent healthcare systems.

It can help doctors diagnose patients in time. This leads to decisions and improved patient outcomes.

Quantum machine learning and deep learning can work together to make healthcare systems smarter.

Doctors can use these systems to make decisions. The integration of quantum machine learning with learning is key to improving patient outcomes.

It enhances the models ability to generalize in challenging scenarios. The model can help doctors diagnose patients accurately. The use of quantum machine learning with learning is a promising step toward next-generation intelligent healthcare systems. It has the potential to assist practitioners, in real-time diagnosis. This can improve decision-making and patient outcomes.

I. Introduction

The use of quantum machine learning with learning not only makes segmentation more precise but also helps the model work well in tough situations. This mix of approaches is a step toward creating intelligent healthcare systems that can help doctors diagnose patients in real-time, which can lead to better clinical decisions and patient outcomes.

Colorectal cancer is a health concern and one of the leading causes of cancer-related deaths worldwide. According to medical reports the number of people getting colorectal cancer is increasing due to changes in lifestyle, diet and aging populations. Finding polyps early through colonoscopy is crucial in reducing deaths as timely treatment can prevent benign growths from becoming malignant tumors. Therefore accurately and efficiently identifying polyps is essential for improving outcomes and reducing the burden on healthcare systems.

- Colonoscopy imaging is considered the way to detect colorectal abnormalities.
- It lets clinicians visually inspect the lining of the colon and identify suspicious lesions.
- However manually inspecting colonoscopy images is demanding.
- Detecting polyps depends heavily on the clinicians expertise and experience.
- The variability in polyp characteristics, such as size, shape, color and texture along with factors like illumination, motion blur and occlusions makes accurate detection difficult.
- Moreover long examination procedures can lead to fatigue increasing the likelihood of missed detections and diagnostic errors.

To address these challenges, automated computer-aided diagnostic (CAD) systems have gained attention in recent years. Advances in intelligence particularly deep learning have revolutionized the field of medical image analysis.

- Convolutional Neural Networks (CNNs) have shown performance in tasks such as classification, detection and segmentation.

- Among architectures U-Net has emerged as a highly effective model for biomedical image segmentation due to its symmetric encoder-decoder structure and skip connections, which enable precise localization and capture of contextual information.

While classical deep learning approaches have achieved success they still face limitations in capturing highly complex feature relationships and generalizing across diverse datasets.

- In this context Quantum Machine Learning (QML) has emerged as a novel and promising paradigm that integrates principles of quantum computing with machine learning.
- QML leverages quantum states and quantum operations to represent and process information in dimensional spaces enabling enhanced feature representation and potentially improved learning efficiency.

In this work we propose a framework that combines the strengths of U-Net and Quantum Machine Learning to improve polyp segmentation performance.

- The U-Net model is employed for extracting features and generating initial segmentation maps.
- The quantum-enhanced layer refines these features by capturing complex patterns that are difficult to model using classical techniques alone.
- The integration of these two approaches aims to achieve segmentation accuracy improved boundary detection and better generalization across challenging medical imaging scenarios.

The proposed framework not only enhances segmentation performance but also contributes toward the development of intelligent automated healthcare systems.

- By reducing dependency on manual analysis the system has the potential to assist medical professionals in real-time diagnosis.
- It can minimize human error and improve clinical decision-making processes.
- This research highlights the growing importance of combining quantum computing techniques for advancing next-generation medical imaging solutions.

II. System Architecture

The system they are talking about has parts that work together. Each part helps make the information it gets a little better. This makes the system better at figuring out what is in the pictures like you can see in Figure 1. The system uses two kinds of machine learning the kind and the quantum kind because each one is good at different things. The system takes pictures from colonoscopies and changes them into very accurate maps of what is in the pictures by doing a series of steps in the right order. The system does this to make the colonoscopy images into accurate segmentation masks. The colonoscopy images go through the system and come out as very good segmentation masks. The system is made to make the segmentation masks very accurate.

- **Input Layer:** The system starts by taking colonoscopy images as input. These images can have resolutions, illumination and quality because of different imaging conditions. The input layer checks that all images are loaded and formatted properly for processing. The colonoscopy images are the key to the process. They are the ones that help the system to work accurately. The quality of colonoscopy images can affect the results. So it is essential to ensure that the colonoscopy images are of quality. The system relies on these colonoscopy images to function correctly.
- **Preprocessing:** The input images go through a lot of changes at this point. They get resized, normalized and cleaned up to get rid of noise. We also do some things to the images, like rotating them, flipping them and scaling them. This helps to make the dataset more diverse. Makes the model better at handling different types of images. The model's generalisation capability is improved by doing this to the input images.
- **U-Net Segmentation:** The images that have been prepared are put through a kind of computer model called U-Net. This U-Net model does two things: it pulls out important details from the images, and it separates the images into different parts. The first part of the U-Net, called the encoder, looks at the images in ways. It uses layers that look at small parts of the image at a time and layers that make the image smaller to understand the images better. The second part, called the decoder, takes what the encoder has learnt and tries to make a map of how the image should be separated. It does this by making the image bigger and using special shortcuts that help it remember where things are in the image. The U-Net model uses these details to create a map that shows the parts of the image. The map is based on the features that the encoder has pulled out and the spatial information that the shortcuts help to preserve. The goal is to get a segmentation map from the U-Net model.
- **Feature Extraction:** High-level feature representations from the U-Net model are refined further. These features capture patterns in space, variations in texture and information about the context of polyp structures in colonoscopy images. The U-Net model helps to identify polyp structures within colonoscopy images. It provides features of polyps. These features are very important for detecting polyps. The features show how polyps look in images. They help doctors to find polyps. Colonoscopy images have textures and patterns. The U-Net model is good at finding these patterns. It helps to refine features for polyp detection. The features are used to improve polyp detection. Polyp structures have patterns. The U-Net model helps to identify these patterns. It provides high-level features of polyps. These features are refined for detection.
- **QML Layer:** The features that we get are then put through a Quantum Machine Learning layer. This is where the classical features are turned into states. We

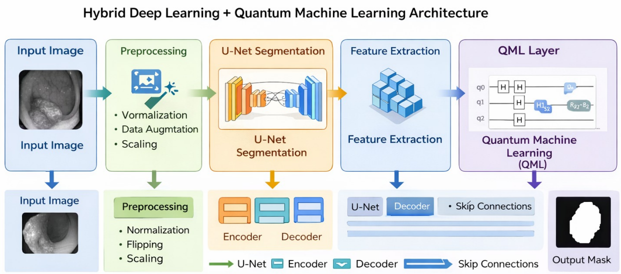

Fig. 1. Proposed hybrid deep learning and QML architecture for polyp segmentation.

also use quantum circuits, with parameters to make the features better. The quantum transformations help the model to see the details and make the segmentation more accurate. We use quantum machine learning to make this happen. It really helps with intricate relationships and precision.

- **Output Layer:** The refined features are then used to create a map that shows where the polyps are. This map is very accurate. It can tell the polyp areas apart from the rest of the picture. The output is like a label that says what each pixel is so we can see the polyp regions very clearly.

III. Methodology

A. Dataset

The colonoscopy images used in this study come with their ground truth segmentation masks. These masks show where the polyps are and what their boundaries are. The images come from medical imaging datasets that are available to the public. Medical experts carefully annotated these images to make sure they are good enough for training.

The dataset has a lot of samples. These samples show polyp sizes and shapes. They also show textures and lighting conditions. This variety is important for building a segmentation model. To train and test the model the dataset is split into three parts. The training set is seventy percent of the dataset. The validation set is fifteen percent. The testing set is also fifteen percent. The training set is used to make the model better. The validation set is used to adjust the model and stop it when it is good enough. The test set is used to see how well the model works.

Splitting the dataset in this way helps the model work with data it has not seen before. This makes the model more general and less likely to be wrong. The colonoscopy images and their ground truth segmentation masks are used to make sure the model is good at finding polyps. The dataset is divided into training, validation and testing sets to make sure the model is working well.

B. Preprocessing

Preprocessing is an important step in the pipeline. It affects the quality of the input data and how well the model training goes. The following things are done to get the data ready:

- **Image Resizing:** All images are changed to a fixed size of 256×256 pixels. This helps in making sure all images have the same size. It also makes it easier to process them together in batches, which reduces complexity.
- **Normalization:** The pixel intensities are changed to be between 0 and 1. This really helps when the computer is learning. It makes the updates to the gradients more stable. The computer can also find the answer it is looking for a lot faster. The pixel intensities being between 0 and 1 is very important for the training process.
- **Noise Reduction:** There are some things that can be seen in colonoscopy images that are not really part of the picture. These things can make it hard to see what is really going on in the image. So we try to get rid of them. We do this so that the important parts of the colonoscopy images are easier to see.
- **Data Augmentation:** To make the data more varied and prevent the model from fitting too closely to the training data we use several methods to increase the data:
  - Horizontal and vertical flipping
  - Rotation at multiple angles
  - Zooming and scaling transformations
  - Random cropping and translation

These preprocessing steps enable the model to learn invariant features and improve robustness against real-world variations.

C. U-Net Architecture

The U-Net architecture is a widely used convolutional neural network specifically designed for biomedical image segmentation tasks. It follows a symmetric encoder-decoder structure that allows the model to capture both contextual and spatial information effectively.

Encoder:

- The encoder path consists of repeated convolutional blocks, each containing convolution layers followed by Rectified Linear Unit (ReLU) activation functions.
- Max-pooling layers are used to progressively reduce spatial dimensions while increasing feature depth.
- This stage captures hierarchical features ranging from low-level edges to high-level semantic representations.

Decoder:

- The decoder path performs upsampling using transpose convolution (deconvolution) layers.
- Feature maps from the encoder are concatenated with corresponding decoder layers through skip connections.
- These skip connections preserve spatial information and improve localization accuracy.

The U-Net architecture effectively balances global context understanding with fine-grained detail preservation, making it highly suitable for segmenting irregular and small structures such as polyps.

D. Quantum Machine Learning Layer

To further enhance the representational power of the model, a Quantum Machine Learning (QML) layer is incorporated into the framework. This layer utilizes variational quantum circuits (VQCs) to process features extracted by the U-Net model.

- **Feature Encoding:** Classical features are encoded into quantum states using encoding techniques such as angle encoding or amplitude encoding.
- **Quantum Transformation:** Parameterized quantum gates (e.g., rotation and entanglement gates) are applied to transform the encoded states.
- **Entanglement:** Quantum entanglement allows interaction between qubits, enabling the model to capture complex feature correlations.
- **Measurement:** The quantum states are measured to obtain classical outputs, which serve as enhanced feature representations.

The QML layer enables the model to explore high-dimensional feature spaces and capture non-linear relationships that are difficult to model using classical neural networks alone. This leads to improved segmentation accuracy and better generalization.

E. Feature Integration

The features extracted from the U-Net model are passed to the quantum layer, where they are transformed and refined. These enhanced features are then integrated back into the segmentation pipeline to improve prediction quality. This hybrid integration ensures that both classical and quantum advantages are effectively utilized.

F. Training Details

The model is trained using a supervised learning approach. The following hyperparameters are used:

- Optimizer: Adam optimizer is used due to its adaptive learning capability.
- Learning Rate: 0.001 for stable convergence.
- Batch Size: 16 to balance memory usage and training efficiency.
- Epochs: 50 to ensure sufficient learning without overfitting.

During training, both training and validation losses are monitored to ensure that the model does not overfit and generalizes well.

G. Loss Function

The Dice loss function is used to optimize segmentation performance. It is particularly suitable for medical image segmentation tasks where class imbalance is common.

TABLE I
SEGMENTATION PERFORMANCE

Metric	Value	Interpretation
Dice	0.9161	Excellent overlap
IoU	0.8701	High similarity
Accuracy	0.9874	Very high

$$\text{Dice} = \frac{2|X \cap Y|}{|X| + |Y|}$$

where X represents the predicted segmentation and Y represents the ground truth mask.

The Dice loss directly measures the overlap between predicted and actual regions, making it more effective than traditional loss functions for segmentation tasks.

IV. Experimental Results

A. Performance Metrics

To evaluate the effectiveness of the proposed model, several performance metrics are used. These metrics provide a comprehensive assessment of segmentation quality:

- **Dice Score:** Measures the overlap between predicted and ground truth masks.
- **Intersection over Union (IoU):** Evaluates the similarity between predicted and actual regions.
- **Pixel Accuracy:** Calculates the proportion of correctly classified pixels.

B. Result Analysis

The experimental results clearly demonstrate that the proposed hybrid U-Net and Quantum Machine Learning (QML) model achieves superior segmentation performance across all evaluation metrics. The Dice score of 0.9161 indicates a high degree of overlap between the predicted segmentation masks and the ground truth annotations, which confirms the model's ability to accurately capture polyp regions.

Similarly, the Intersection over Union (IoU) score of 0.8701 reflects strong agreement between predicted and actual regions, highlighting the model's effectiveness in minimizing both false positives and false negatives. The pixel accuracy of 0.9874 further validates that the majority of pixels are correctly classified, demonstrating the reliability of the model in pixel-wise segmentation tasks.

The integration of the Quantum Machine Learning layer plays a significant role in enhancing feature representation. By leveraging quantum transformations, the model is able to capture complex non-linear relationships within the data, which improves boundary delineation and segmentation precision. This is particularly beneficial in challenging scenarios where polyps exhibit irregular shapes, low contrast, or blurred boundaries.

Additionally, the model shows consistent performance across diverse samples, indicating strong generalization capability. The hybrid architecture effectively combines the

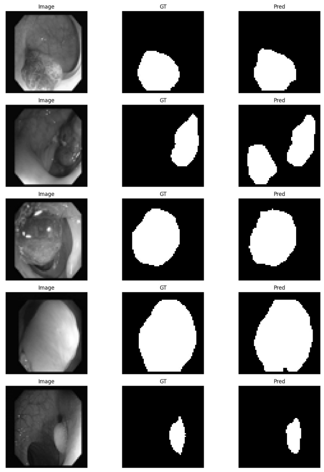

Fig. 2. Input Image, Ground Truth, and Predicted Segmentation

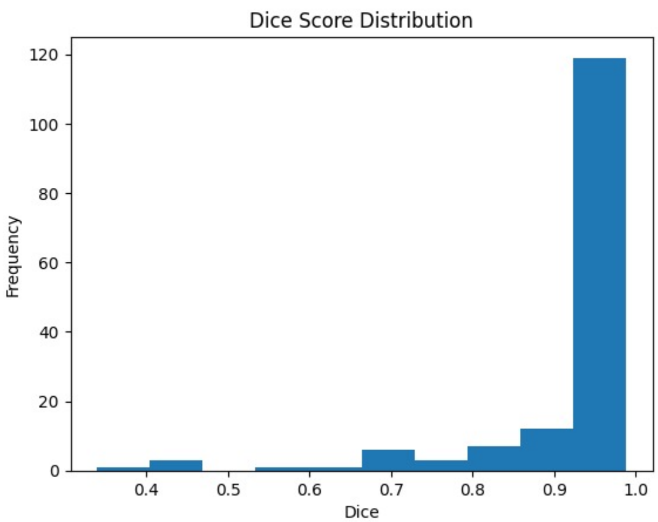

Fig. 3. Distribution of Dice scores over the evaluated samples.

strengths of classical deep learning and quantum computing, resulting in improved robustness and accuracy.

C. Visualization Results

The visualization results provide qualitative evidence of the model's performance. As shown in Fig. **??**, the predicted segmentation closely matches the ground truth mask, accurately identifying the polyp region with well-defined boundaries. The model successfully distinguishes between polyp and background regions, even in cases with complex textures and varying illumination conditions.

D. Performance Analysis

The performance curves, including Dice score and IoU score per batch, indicate stable training behavior with gradual convergence. Minor fluctuations are observed during training, which can be attributed to challenging samples with irregular polyp structures and varying contrast levels.

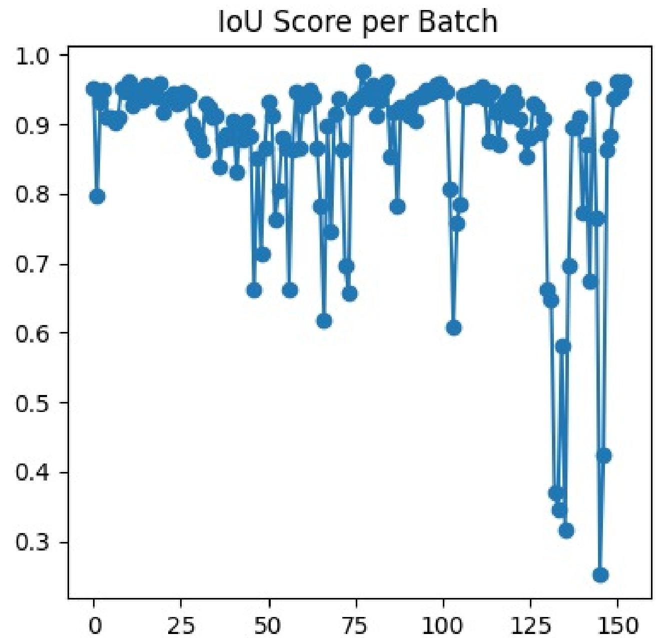

Fig. 4. IoU score per batch during evaluation.

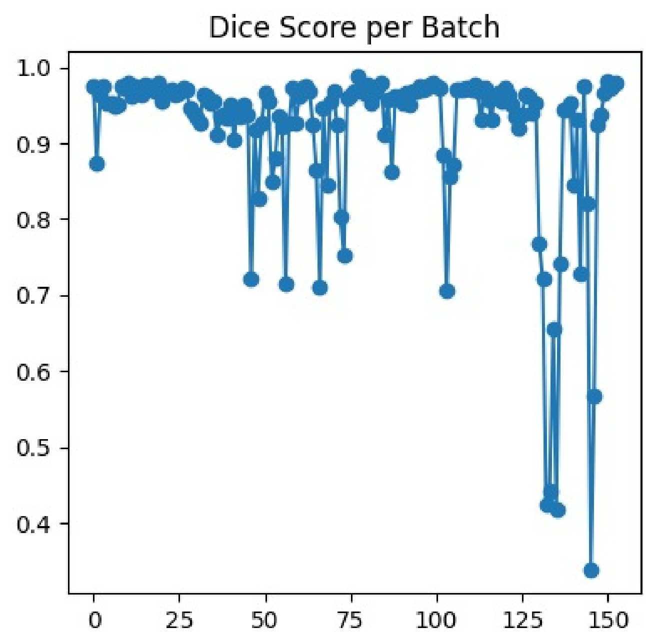

Fig. 5. Dice score per batch during evaluation.

Despite these variations, the overall trend shows consistent improvement, indicating effective learning and optimization. Pixel accuracy remains consistently high throughout training, suggesting that the model maintains reliable performance across different batches.

Furthermore, the combination of data augmentation and the hybrid architecture contributes to improved generalization and prevents overfitting.

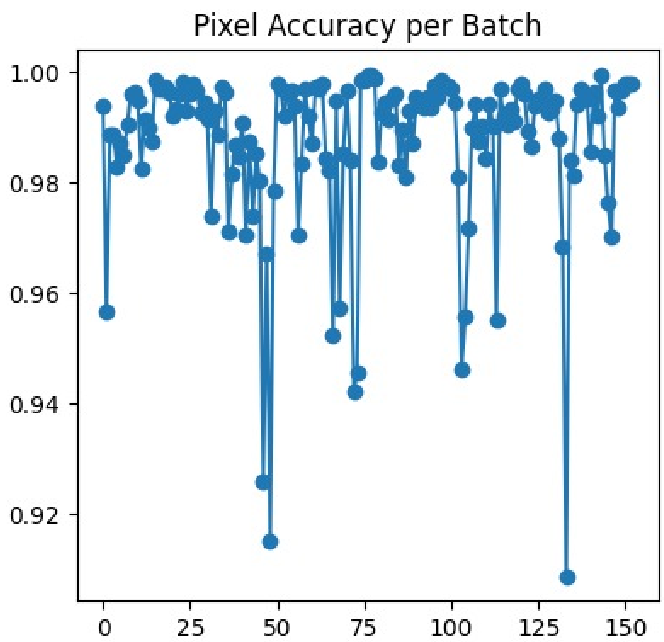

Fig. 6. Pixel accuracy per batch during evaluation.

E. Discussion

The results demonstrate that the proposed hybrid model achieves a balance between accuracy and robustness. The U-Net architecture effectively captures spatial and contextual features, while the QML layer enhances feature representation by exploring high-dimensional quantum feature spaces.

This combination leads to improved boundary detection and segmentation precision, especially in challenging cases where traditional models struggle. Compared to conventional deep learning approaches, the hybrid model provides better performance and improved adaptability.

The findings suggest that integrating quantum machine learning with deep learning architectures has significant potential for advancing medical image analysis and developing intelligent diagnostic systems.

V. Comparative Analysis

To further evaluate the effectiveness of the proposed model, a comparative analysis is conducted with baseline models such as conventional CNN and standard U-Net architectures.

TABLE II
Comparison with Existing Models

Model	Dice	IoU	Accuracy
CNN	0.82	0.75	0.95
U-Net	0.89	0.83	0.97
Proposed Model	**0.9161**	**0.8701**	**0.9874**

The results clearly show that the proposed hybrid model outperforms traditional methods across all evaluation metrics. The improvement in Dice and IoU scores indicates better segmentation quality and more accurate region detection.

The CNN model, while effective for basic feature extraction, lacks the ability to capture fine-grained spatial details. The standard U-Net model improves segmentation performance through its encoder-decoder structure, but still has limitations in modeling complex feature relationships.

In contrast, the proposed hybrid model leverages the strengths of both U-Net and Quantum Machine Learning, resulting in enhanced feature representation and improved segmentation accuracy. The incorporation of the QML layer enables the model to capture intricate patterns and relationships, leading to superior performance.

Overall, the comparative analysis validates the effectiveness of the proposed approach and demonstrates its advantage over existing methods in medical image segmentation tasks.

VI. Advantages

The proposed hybrid framework offers several significant advantages over traditional segmentation approaches by effectively combining deep learning and quantum machine learning techniques:

- **High Segmentation Accuracy:** The model achieves superior performance in terms of Dice score, IoU, and pixel accuracy, indicating a high degree of overlap between predicted and ground truth masks. This ensures precise detection of polyp regions, even in cases with complex structures and subtle boundaries.
- **Robust Performance Under Variations:** Through the use of extensive data augmentation and hierarchical feature extraction, the model demonstrates robustness to variations in illumination, noise, scale, and polyp morphology. This makes it suitable for handling real-world colonoscopy data with diverse imaging conditions.
- **Enhanced Feature Representation Using QML:** The integration of Quantum Machine Learning introduces an additional layer of feature transformation, enabling the model to capture complex, non-linear relationships within the data. This leads to improved segmentation quality compared to purely classical approaches.
- **Improved Boundary Detection:** The combination of U-Net skip connections and quantum-enhanced features allows for precise delineation of polyp boundaries. This is particularly important in medical imaging, where accurate boundary detection directly impacts diagnostic quality.
- **Better Generalization Capability:** The hybrid architecture, along with data augmentation strategies, enables the model to generalize well across unseen data. This reduces the risk of overfitting and improves performance on diverse datasets.
- **Reduced Human Dependency:** The automated segmentation process significantly reduces reliance on manual analysis by clinicians. This helps minimize human error, improves consistency, and reduces the workload on medical professionals.

- **Scalability for Advanced Applications:** The modular design of the framework allows easy integration with advanced architectures such as Attention U-Net or transformer-based models, making it adaptable for future enhancements.
- **Potential for Real-Time Clinical Deployment:** With appropriate optimization and hardware acceleration, the model can be deployed in real-time clinical environments to assist doctors during colonoscopy procedures, enabling faster and more accurate diagnosis.
- **Foundation for Next-Generation Healthcare Systems:** The integration of quantum machine learning positions this work as a step toward next-generation intelligent healthcare systems, where classical and quantum technologies work together to improve medical decision-making.

VII. Limitations

Despite its strong performance, the proposed hybrid deep learning and quantum machine learning model has several limitations that need to be considered for practical deployment and further research:

- **High Computational Complexity:** The integration of deep learning architectures such as U-Net with quantum simulation layers significantly increases computational requirements. Training the model involves processing high-dimensional image data along with quantum circuit evaluations, which results in increased training time and higher memory consumption.
- **Quantum Simulation Overhead:** Due to the current limitations of quantum hardware, the Quantum Machine Learning (QML) component is implemented using classical simulation frameworks. This introduces additional computational overhead and may not fully exploit the theoretical advantages of quantum computing, such as exponential speedup.
- **Dependency on Dataset Quality:** The performance of the model is highly dependent on the quality, diversity, and annotation accuracy of the dataset. Limited datasets or biased samples may lead to overfitting and reduced generalization capability when applied to real-world clinical scenarios.
- **Scalability Challenges:** Scaling the proposed model to handle large-scale datasets or real-time applications can be challenging. The combination of deep neural networks and quantum layers requires substantial computational resources, which may not be readily available in standard clinical environments.
- **Complex Implementation:** The integration of classical deep learning models with quantum circuits requires interdisciplinary expertise in both machine learning and quantum computing. This increases the complexity of implementation, debugging, and optimization compared to conventional models.
- **Limited Interpretability:** Like many deep learning models, the proposed framework operates as a black-box system, making it difficult to interpret the decision-making process. This lack of interpretability may limit its acceptance in clinical settings where explainability is crucial.
- **Hardware Constraints:** Current quantum devices are still in the early stages of development and suffer from issues such as noise, decoherence, and limited qubit availability. These constraints restrict the practical implementation of QML components on real quantum hardware.
- **Inference Latency:** The addition of quantum layers may increase inference time, which could be a limitation for real-time applications such as live colonoscopy assistance unless optimized.

VIII. Future Work

Future research can focus on improving and extending the proposed framework in several directions:

- **Real-Time Clinical Deployment:** Optimizing the model for real-time inference to assist doctors during live colonoscopy procedures.
- **Integration of Advanced Architectures:** Incorporating advanced segmentation models such as Attention U-Net or DeepLabV3+ to further enhance performance.
- **Implementation on Real Quantum Hardware:** Deploying the QML component on actual quantum devices to fully leverage quantum advantages.
- **Multimodal Data Integration:** Combining colonoscopy images with other medical data such as patient history or biopsy results to improve diagnostic accuracy.
- **Improved Data Augmentation Techniques:** Using advanced augmentation strategies and synthetic data generation to enhance model robustness.
- **Optimization for Edge Devices:** Developing lightweight versions of the model for deployment on portable medical devices.

IX. Conclusion

In this paper, a hybrid deep learning and quantum machine learning framework for polyp segmentation in colonoscopy images is presented. The proposed approach leverages the strengths of U-Net for feature extraction and segmentation, along with quantum-enhanced learning for improved feature representation.

Experimental results demonstrate that the model achieves high performance across multiple evaluation metrics, including Dice score, IoU, and pixel accuracy. The integration of quantum machine learning contributes to better boundary detection and improved segmentation precision, particularly in challenging cases with complex polyp structures.

The proposed framework highlights the potential of combining classical and quantum approaches for advancing medical image analysis. By enabling accurate and automated polyp detection, the system can assist healthcare professionals in early diagnosis and improve patient outcomes. This work represents a significant step toward the development of intelligent, next-generation computer-aided diagnostic systems.

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